Project Report

(Number Guessing Game)

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Language: Python 3

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1. Project Overview

The **Number Guessing Game** is a game where the computer generates a random number within a defined range (e.g., 1 to 100), and the player attempts to guess it. The program provides feedback on whether the guess is "Too High" or "Too Low" until the player finds the correct number.

2. Objectives

- To implement basic programming constructs: **loops, conditionals, and variables**.
- To utilize the Python random module for dynamic output.
- To practice **input validation** and exception handling (preventing crashes if a user types a letter instead of a number).

3. System Architecture & Logic

The game follows a linear logic flow, often referred to as a "While-Loop" structure.

Key Components:

- **Randomization:** Uses random.randint(a, b) to ensure every game is unique.

- **Comparison Logic:** A series of if-elif-else statements to guide the user.
- **Score Tracking:** A counter variable that increments with every incorrect guess to show the user their "efficiency" at the end.

4. Features

- **User-Friendly Feedback:** Immediate hints based on user input.
- **Input Handling:** Ensures the game continues even if the user enters invalid data.
- **Replayability:** Option to restart the game without exiting the script.

5. Technical Snippets

The core logic relies on the following mathematical comparison:

$$| \text{Target} - \text{Guess} | = 0 \implies \text{Win}$$

If the difference is positive or negative, the program triggers the "Too High" or "Too Low" strings respectively.